

Progressive Systemic Sclerosis

Agnes Fogo, MD

The AJKD Atlas of Renal Pathology presents a compilation of figures on a specific pathologic entity. You may download the figures to create your own personal, non-commercial library of images or to create slides for teaching purposes.

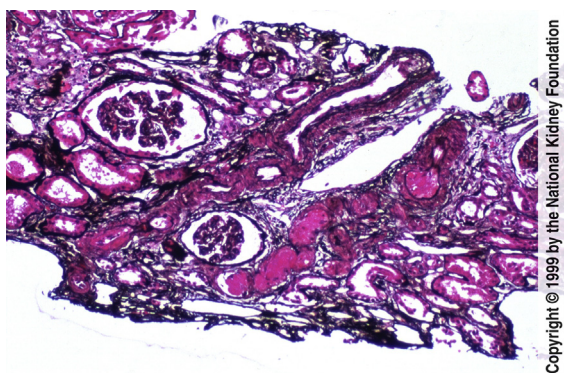


Fig 1. In progressive systemic sclerosis, many features overlap with those seen in malignant hypertension, with fibrinoid necrosis of vessels. There may also be fibrin thrombi extending into glomeruli, overlapping with the lesions typical of thrombotic microangiopathy. However, in typical progressive systemic sclerosis, injury often extends to larger vessels, contrasting the typical predominance of glomerular involvement in thrombotic microangiopathy. An uninvolved interlobular artery and an interlobular artery with extensive fibrinoid necrosis are shown in longitudinal section, with only ischemic changes in glomeruli. (Jones' silver stain, original magnification $\times 200$).

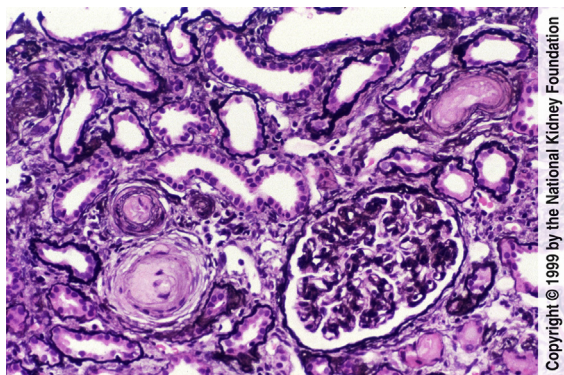


Fig 2. The artery shows early organization with "onion skin" change caused by lamellation and mucoid change with swelling of the intimal layer, with corrugation of the glomerular basement membrane, in this case of progressive systemic sclerosis. (Jones' silver stain, original magnification $\times 200$).

From the Department of Pathology, Vanderbilt University Medical Center, Nashville, TN.

Medical Photographer: Brent Weedman; with assistance from Kim Soletz, MD, of the National Kidney Foundation's cyberNephrology™ Team.

Address author queries to Agnes Fogo, MD, Department of Pathology, Vanderbilt University Medical Center, MCN C-3310, Nashville, TN 37232. E-mail: Agnes.Fogo@vanderbilt.edu

Am J Kidney Dis. 34(2):E4-E5.

© 1999 by the National Kidney Foundation, Inc. Published by Elsevier Inc. All rights reserved.

0272-6386/\$36.00

[http://dx.doi.org/10.1053/S0272-6386\(13\)90019-6](http://dx.doi.org/10.1053/S0272-6386(13)90019-6)

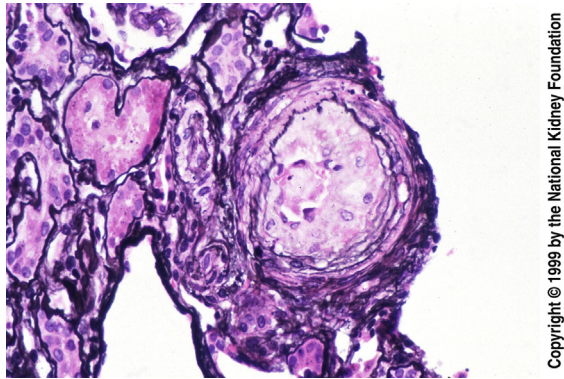


Fig 3. In early stages of progressive systemic sclerosis, mucoid changes in the vessel wall predominate, with swollen endothelium and a mucoid appearance of the subintima, frequently with red blood cell fragments, as seen here. (Jones' silver stain, original magnification $\times 400$).

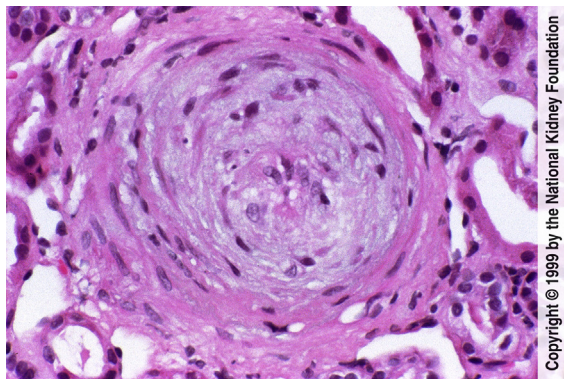


Fig 4. In the more chronic stage of progressive systemic sclerosis, fibrous organization of the intimal injury results in a lamellated "onion skin" appearance of arteries. (Periodic acid Schiff, original magnification $\times 400$).

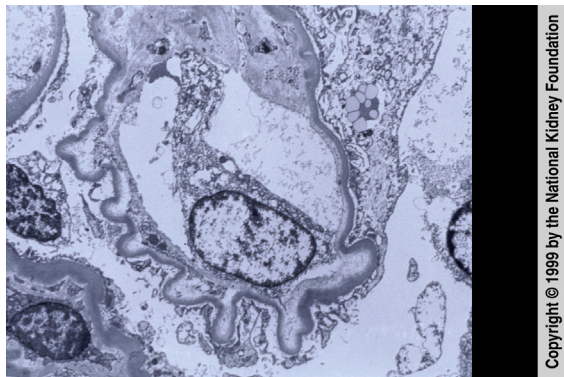


Fig 5. The glomeruli show evidence of endothelial injury and ischemia, visualized by corrugation and expansion of the lamina rara interna in this case of progressive systemic sclerosis. (Transmission electron microscopy, original magnification $\times 3,000$).